Unit 3

Linked list

1) In a singly linked list, the link field of the last node generally contains: (1 Mark)
A) Address of the first node
B) Address of the previous node
C) NULL
D) Address of the root node

2) Analyze the role of Node, Data field, Next pointer, and NULL pointer in the representation of a singly linked list. Explain what happens if the Next pointer of a node is incorrectly assigned (5 Mark)

3) Write a C program to implement a singly linked list with functions for the following operations: (5 Mark)
a) Insertion at the beginning
b) Insertion at the end
c) Deletion from the beginning
d) Display the linked list

4) Which linked list contains a link from the last node back to the first node? (1 Mark)
A) Singly Linked List
B) Doubly Linked List
C) Circular Linked List
D) Linear List

5) Analyze the time complexity of the following singly linked-list operations: (5 Mark)
a) Creating a linked list
b) Traversing the list
c) Searching for an element
d) Inserting at the beginning
e) Deleting a node

6) Write a C program to create a doubly linked list and perform forward and backward traversal. Evaluate the advantages of maintaining both prev and next pointers. (5 Mark)

7) In a doubly linked list, each node normally contains: (5 Mark)
A) Data and one pointer
B) Data and two pointers
C) Only two pointers
D) Data and three pointers

8) What is a linked list? Explain the different types of linked lists with suitable diagrams. (5 Mark)